

Sebastian Gracia Guzman

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PROFESSIONAL SUMMARY

Data scientist with a master's in mathematics and 3+ years of industry experience developing predictive models and machine learning solutions that drive growth and operational efficiency at scale. Proficient in Python, SQL, and modern ML libraries including PyTorch. Skilled at collaborating across technical and non-technical teams to drive tangible business value.

KEY SKILLS

Statistics / ML:	Predictive Modeling, Causal Inference, Demand Forecasting, Customer Segmentation, Time Series Analysis, A/B Testing, Experiment Design, Reinforcement Learning, Optimization, Pricing Elasticity
Programming:	Python, R, SQL, MATLAB, Java
Packages:	PyTorch, NumPy, Pandas, SciPy, Scikit-Learn, XGBoost, OpenAI, PySpark, Databricks
Visualization:	Tableau, Power BI, matplotlib, seaborn, ggplot, Executive Level Presentations

PROFESSIONAL EXPERIENCE

Data Scientist

Nov 2021 – Jan 2025

Primrose Schools Franchising Company | Atlanta, Georgia

- Developed a hierarchical mixed effects model analyzing 500+ franchise locations identifying a \$25M revenue optimization opportunity through facility reinvestment
- Led end-to-end execution of 15+ analytics projects — from ingesting raw data across multiple sources and conducting exploratory analysis to developing predictive models, presenting insights to stakeholders, and deploying results in executive dashboards
- Applied Gaussian Mixture Models for customer LTV segmentation, discovering key retention drivers (e.g., sibling enrollment extending tenure by 1.5 years) that reshaped customer targeting approaches
- Implemented Git-based model versioning and systematic retraining processes for reproducible, production-grade analytics
- Delivered quarterly presentations to C-suite executives, translating complex analyses into actionable insights
- Promoted to Senior Data Scientist in 2023 based on consistent delivery of high-impact models and learnings

Graduate Teaching Instructor

Aug 2019 - May 2021

Clemson University | Clemson, South Carolina

- Employed clear communication skills to achieve an exam average ~6 points higher than other Calculus sections

PROJECTS

Master's Research: Policy Gradient Methods in Deep Reinforcement Learning

Investigated the mathematical theory behind policy-based RL methods, covering MDPs, value functions, and the Policy Gradient Theorem. Implemented and trained agents using PyTorch across multiple environments (HalfCheetah, Pong). Separately implemented Rainbow DQN for autonomous driving, combining multiple DQN variants to replicate benchmark results from the literature.

Predicting Housing Sale Price Intervals — Kaggle

Built a PyTorch neural network trained with Winkler loss to predict calibrated price intervals rather than point estimates. Achieved 93.9% empirical coverage exceeding the 90% nominal target through feature engineering and architectural experimentation.

The Ultimate Statistical Fighter: UFC Fighter Segmentation

Engineered a dataset of 6K+ UFC fights with 100+ features (striking, grappling, physiological). Applied K-Means + PCA to identify distinct fighting archetypes and quantify stylistic matchup advantages across weight classes.

EDUCATION

Master of Science (M.S), Mathematics, Concentration in Operations Research

May 2021

Clemson University | Clemson, South Carolina | GPA: 3.7/4.0

Bachelor of Science (B.S), Mathematics, Minor in Music

May 2019

University of Portland | Portland, Oregon

CERTIFICATIONS

MLOps Essentials: Model Development and Integration, Deployment and Monitoring — *LinkedIn Learning*